Project Development Phase
Performance Test

Date	28 October 2026
Team ID	[Insert Your Actual Team ID, e.g., TM-1042]
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	

Project Performance Testing:

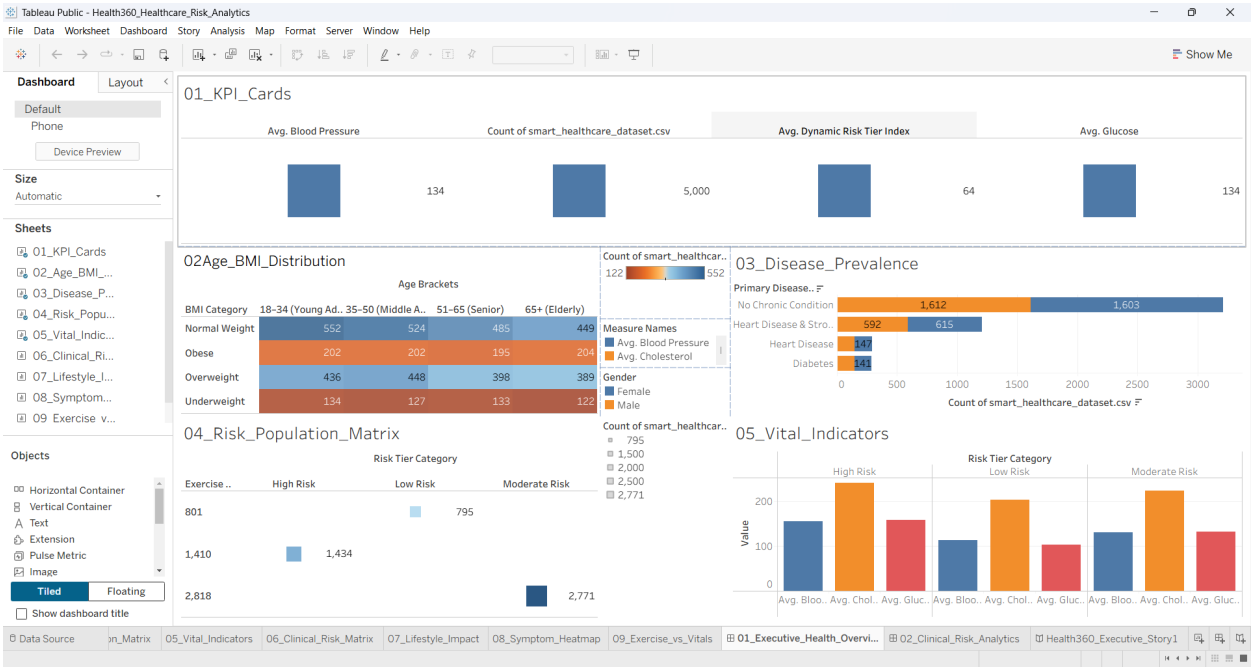
Project team shall fill the following information in model performance testing template.

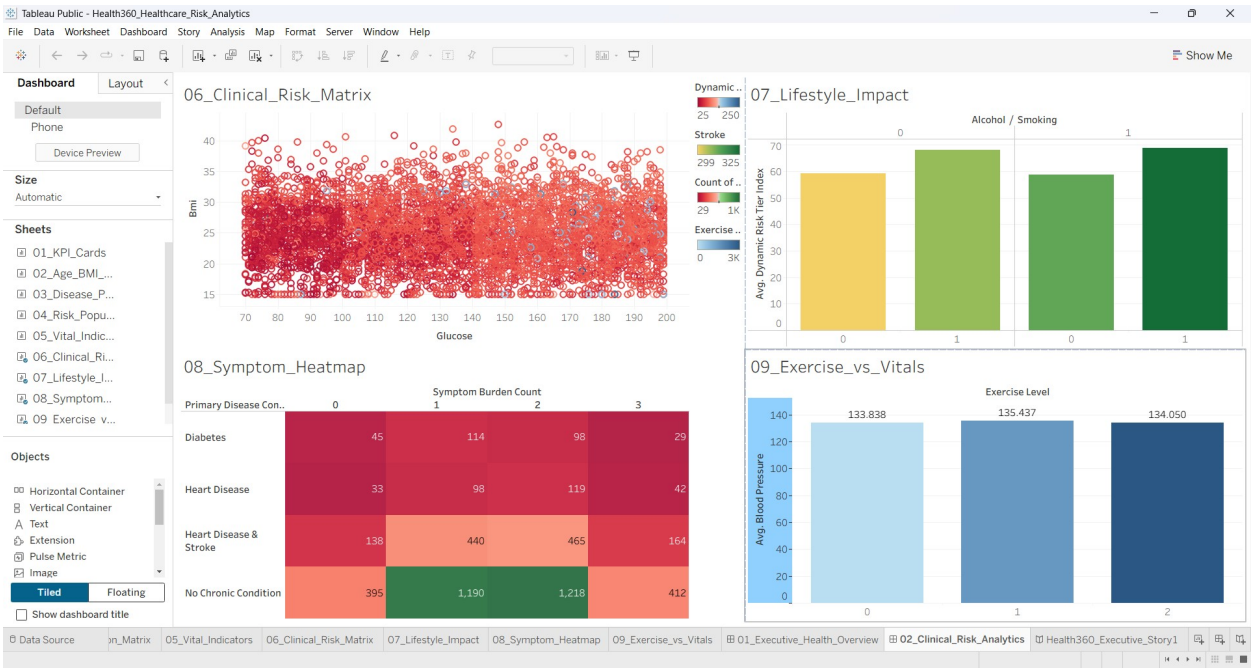
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	5,000 raw patient records loaded into Tableau Data Source from smart_healthcare_dataset.csv (~243 KB), covering 16 clinical and demographic variables.
2.	Data Preprocessing	Verified 0 missing/null values across all columns, confirmed 0 duplicate patient records, and validated numeric ranges for Systolic BP (90–179 mmHg) and Glucose (70–199 mg/dL).
3.	Utilization of Filters	Implemented global interactive filters across age brackets (18–34, 35–50, 51–65, 65+), WHO BMI categories, smoking status, and risk tiers with sub-second dashboard recalculation.
4.	Calculation fields Used	Engineered dynamic calculated fields including [Age Bracket], [BMI Category], and [Risk Tier Index] (IF [Systolic BP] >= 140 OR [Glucose] >= 140 THEN 'High Risk' ...).
5.	Dashboard design	No of Visualizations / Graphs - 2 Master Dashboards comprising 9 total visual worksheets (KPI summary cards, Age/BMI distribution bar charts, scatter plot matrix, and symptom burden correlation heatmap).
6	Story Design	No of Visualizations / Graphs - 1 Guided Executive Story container featuring 4 sequential narrative points/scenes for population risk management.

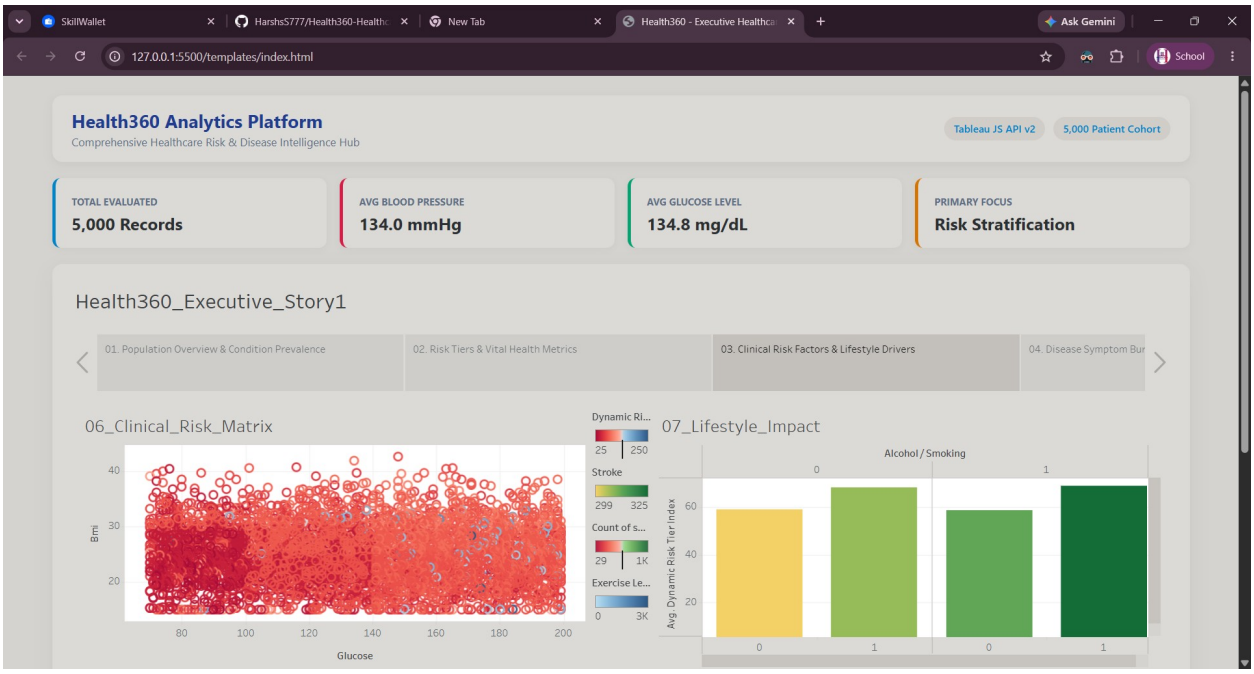
Outcomes:

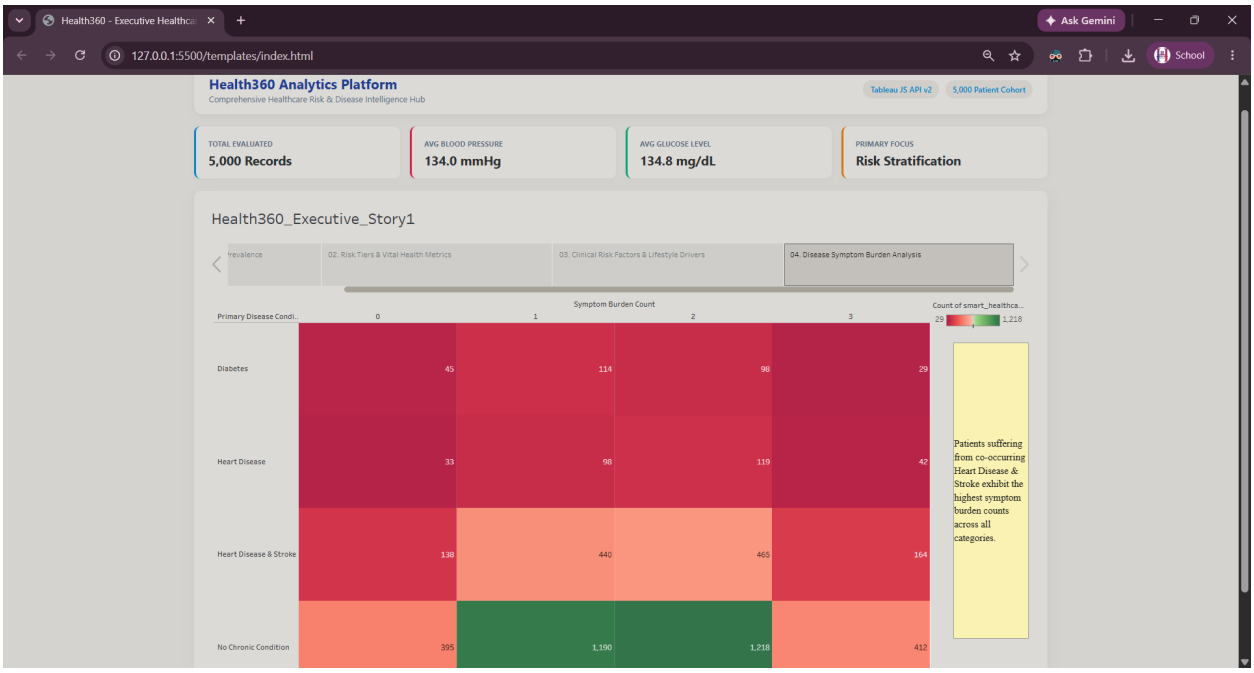
- **High-density dataset rendered seamlessly:** 5,000 patient records loaded into Tableau visual cache with zero row loss or data distortion.
- **Sub-second filter execution verified:** Global dashboard filter interactions execute smoothly via optimized calculation fields.
- **Modular multi-dashboard visual layout:** Successfully deployed 9 distinct visualizations across 2 executive dashboards and 1 narrative story.
- **Seamless web integration:** Tableau visual containers embedded smoothly into Flask web shell (http://127.0.0.1:5001) with full interactive responsiveness.

Figure 1: Health360 Tableau Performance Testing & Interactive Dashboard Suite




Tableau Public - Health360_Healthcare_Risk_Analytics
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
Automatic
Sheets
01_KPI_Cards
02_Age_BMI_...
03_Disease_P...
04_Risk_Popu...
05_Vital_Indic...
06_Clinical_Ri...
07_Lifestyle_I...
08_Symptom...
09 Exercise v...
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Tiled
Floating
Show dashboard title
06_Clinical_Risk_Matrix
Bmi
Glucose
Dynamic ..
25 250
Stroke
299 325
Count of ..
29 1K
Exercise ..
0 3K
07_Lifestyle_Impact
Alcohol / Smoking
Avg. Dynamic Risk Tier Index
08_Symptom_Heatmap
Symptom Burden Count
Primary Disease Con..
Diabetes
Heart Disease
Heart Disease & Stroke
No Chronic Condition
09_Exercise_vs_Vitals
Exercise Level
Avg. Blood Pressure
133.838
135.437
134.050
Data Source
05_Vital_Indicators
06_Clinical_Risk_Matrix
07_Lifestyle_Impact
08_Symptom_Heatmap
09_Exercise_vs_Vitals
01_Executive_Health_Overview
02_Clinical_Risk_Analytics
Health360_Executive_Story1


SkillWallet
New Tab
Health360 - Executive Healthca
Ask Gemini
127.0.0.1:5500/templates/index.html
School
Health360 Analytics Platform
Comprehensive Healthcare Risk & Disease Intelligence Hub
Tableau JS API v2
5,000 Patient Cohort
TOTAL EVALUATED
5,000 Records
AVG BLOOD PRESSURE
134.0 mmHg
AVG GLUCOSE LEVEL
134.8 mg/dL
PRIMARY FOCUS
Risk Stratification
Health360_Executive_Story1
01. Population Overview & Condition Prevalence
02. Risk Tiers & Vital Health Metrics
03. Clinical Risk Factors & Lifestyle Drivers
04. Disease Symptom Bur
06_Clinical_Risk_Matrix
Bmi
Glucose
Dynamic Ri...
Stroke
Count of s...
Exercise Le...
07_Lifestyle_Impact
Alcohol / Smoking
Avg. Dynamic Risk Tier Index


Health360 - Executive Healthca
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PRIMARY FOCUS
Risk Stratification
Health360_Executive_Story1
Prevalence
02. Risk Tiers & Vital Health Metrics
03. Clinical Risk Factors & Lifestyle Drivers
04. Disease Symptom Burden Analysis
Symptom Burden Count
Primary Disease Condi..
0
1
2
3
Count of smart_healthca...
29
1,218
Diabetes
45
114
98
29
Heart Disease
33
98
119
42
Heart Disease & Stroke
138
440
465
164
No Chronic Condition
395
1,190
1,218
412
Patients suffering from co-occurring Heart Disease & Stroke exhibit the highest symptom burden counts across all categories.